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A computational study on nanofluid impingement jets in thermal management of photovoltaic panel

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ABSTRACT

Thermal management is essential to improve the overall performance of photovoltaic (PV) cells. This study presents a comprehensive numerical study on a PV panel to achieve economic efficiency cooling system. A nanofluid jet impingement cooling (JIC) system with different configurations is developed and integrated into the PV panel to control the surface temperature. A three-phase mixture model shows a satisfactory agreement between the simulation and measurement values. Then, the simulations are conducted to understand the importance of different influencing parameters, including inlet temperature (20–40 °C), number of nozzles (8–24), mass flow rate (0.045–0.12 kg/s), jet-to-surface distance (5.1–55 mm), and nozzle diameter (1 and 2 mm). Numerical results demonstrate that uncooled PV temperature reaches 68.5 °C, while using the JIC system reduces that by 36.4 °C. Lower inlet temperatures and larger mass flow rates result in higher PV efficiency. The number of nozzles affects the surface temperature and its uniformity. Small jet-to-surface distances enhance the turbulence kinetic energy and the heat transfer rate. For the nozzle diameters of 1 and 2 mm, the maximum power increment increases to 20.36% and 20.19%, respectively. However, the nozzle diameter of 1 mm increases the pumping power with a coefficient of energy of 1.18.

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1. Introduction

Despite significant efforts, fossil fuels are still the primary energy source that has negatively impacted the environment due to their contribution to green gaseous emission affecting global warming [1]. Among different renewable energy solutions, solar energy is a sustainable resource and cost-competitive alternative with low environmental impact, drawing significant attention from academia and industry [2]. It is also an excellent alternative to tackle the shortages of fossil fuels. Photovoltaic (PV) panels are widely used to absorb solar irradiation and convert it to electricity. However, the existing data confirms that the efficiency of PV modules decreases as the temperature of PV cells increases [3]. Different design and environmental parameters affect the PV performance, and hence previous studies have proposed various techniques to improve the productivity of PV modules [4–6]. It was demonstrated that, compared to the uncooled PV module, the electrical efficiency was enhanced by 38.4% [7] to 80% [8], depending on the effectiveness of cooling techniques.

Among different techniques, jet impingement cooling (JIC) effectively controls the PV temperature and temperature uniformity, consequently enhancing the PV efficiency [2]. Nadda et al. [9] reviewed different studies on the use of impingement jets in PV modules and solar collectors. Bahaidarah [10] studied the performance of a PV module, including a jet impingement cooling system. It was shown that adding the JIC system reduced the maximum temperature by 33 °C and increased the efficiency to 82.6%. Javidan and Jabari [2] examined different arrays of water impinging jets to provide an effective cooling system. Under the optimum condition, the output power was enhanced by 47.67%. It was also found that the temperature distribution was more uniformed at smaller jet-to-surface spacing and higher mass flow rates.

The combination of nanofluids and jet impingement could further improve the cooling performance [11]. It was reported that nanofluid impinging jets enhanced the electrical output power and thermal efficiency more than water in a PV/JIC system [12]. Sun et al. [13] studied the nanofluid heat transfer of a single impinging jet. They showed that heat transfer caused by a circular jet was

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Nomenclature

Latin symbols

A	(m^2) Area of the PV panel
Cp	(J/(K•kg)) Specific heat
H	(mm) Jet-to-surface distance
I	(W/m^2) Solar irradiation
k	(W/(m•K)) Thermal conductivity
$\dot{m}$	(kg/s) Mass flow rate
N	Nozzle number
P	(Pa) Pressure
P	(W) Power
t	(s) Time
u, v, w	(m/s) Velocity
x, y, z	(m) Cartesian coordinates

Greek symbols

dn	(mm) Nozzle diameter
φ	(%) Nanoparticle concentration
ρ	(Kg/m^3) Density
η	(%) Efficiency
ε	Emissivity
α	Absorptivity
τ	Transmissivity

Subscripts

in	Inlet
m	Mixture
nf	Nanofluid

higher than a square jet. Sangeetha et al. [14] studied the effect of different nanofluids on the PV cooling system. They concluded that the electrical efficiency was enhanced by 52, 55, and 60% by adding CuO, Al_2O_3, and multiwall carbon nanotube (MWCNT) particles, respectively. Further study confirms that the Al_2O_3-water nanofluid was more effective than the TiO_2-water nanofluid to enhance the PV electrical efficiency [15]. Later, Radwan and Ahmed [16] found that SiC particles had even better performance than Al_2O_3 nanoparticles to improve electrical efficiency. SiC-water nanofluids with a concentration of 4% resulted in a temperature reduction of 3–8 °C. Rejeb et al. [17] compared different nanoparticles in a PV cooling system and found the best performance using Cu-water nanofluids.

Despite previous experimental studies on PV/JIC, there is a lack of detailed information on how the impingement jets affect the performance of PV modules. This study focuses on the use of computational simulations to develop an effective cooling model that improves the performance of PV modules. To the best of the authors' knowledge, this is the first numerical analysis on the use of multiple impinging jet cooling systems in PV modules. The proposed model in this study is designed based on nanofluid jet impingement to enhance electrical output power and efficiency. A mixture model of three phases (air, water, and SiC particles) is adopted to solve governing equations in the fluid zone. This study provides new analysis on the importance of key influencing parameters, including the inlet temperature (20–40 °C), the number of nozzles (8–24), mass flow rate (0.045–0.12 kg/s), jet-to-surface spacing (5.1–55 mm), and nozzle diameter (1 and 2 mm). Different performance parameters, including PV surface temperature and uniformity, PV efficiency, and the maximum power increment percentage ($\%P_{(\max)increase}$), are used to explore an effective cooling system. In addition, the economic aspect of the output power enhancement is deeply analysed by a coefficient of energy (*COE*) that will provide new insights for the effective design of PV/JIC systems.

2. Methodology

2.1. Computational domain and geometry description

A 3D model is developed for a PV module with JIC. Fig. 1 illustrates a schematic diagram of the PV/JIC system, which is consistent with Javidan and Moghadam's experimental study [2]. According to the existing datasheet, the PV module model of ZT10-18-P consists of 72 cells with 24 × 36 mm dimensions. The overall dimensions of the PV module (L × W × t) are 350 × 270 × 17 mm [2]. Its efficiency and maximum power are 16.5% and 10 W, respectively. The nominal operating cell temperature (NOCT) is 45 ± 2 °C. The thermophysical and geometrical properties of different layers in the PV module are listed in Table 1.

The thermophysical properties of pure water and SiC nanoparticles are shown in Table 2. Based on previous studies, SiC nanoparticles have better performance than other materials, and hence they are considered in this study [16,17]. SiC nanoparticles at a concentration of 4% are added to the water as the sedimentation is negligible for small nanoparticles and lower concentrations [20,21].

The total number of nozzles equals 24 while the effect of different IJ numbers (8, 12, 16, 20, and 24) is examined on the JIC system by shutting off some nozzles, as shown in Fig. 2. The effect of jet-to-surface distance (which is the same as *H* shown in Fig. 1(b)) on the JIC system is analysed for three different values: 5.1, 30, and 55 mm. Further analysis is also conducted on the effect of the nozzle size by modelling nozzles with the diameter of 1 and 2 mm.

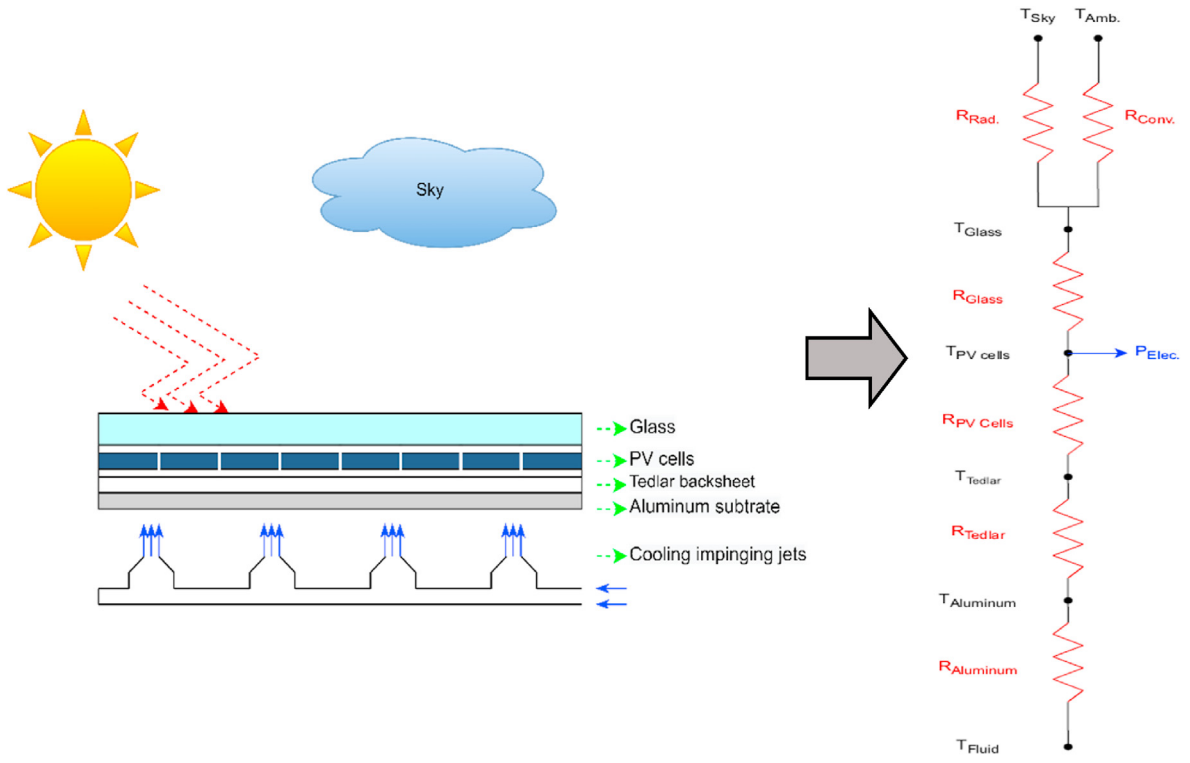
2.2. Assumptions

The following assumptions are considered for the simulations in this study.

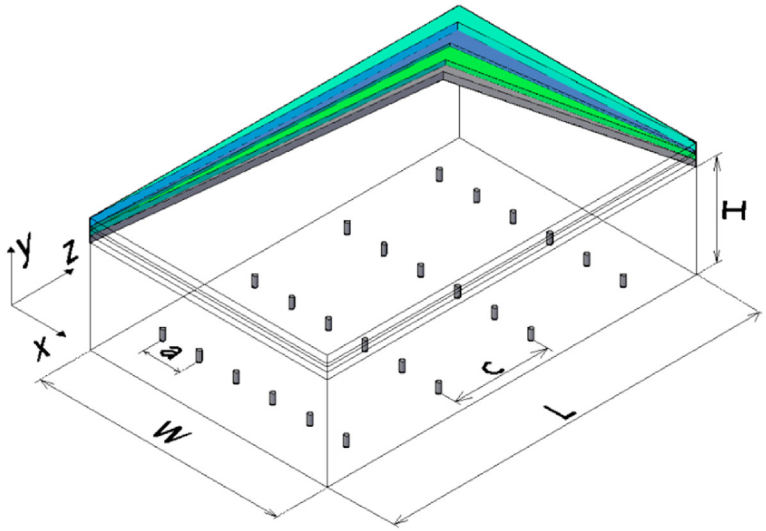
- A particle concentration of 4% is added to the water.
- The coolant is assumed incompressible and Newtonian.
- A solar irradiation level of 1000 W/m^2 is uniformly applied [23].
- The sky temperature is considered a black body's temperature [24].
- Thermal resistance is neglected between different layers [25].
- The SiC nanoparticles are spherical with a diameter of 20 nm.
- The metal frame and top ethylene vinyl acetate (EVA) layer are negligible in numerical simulations due to their small thickness [26–29] to save computational costs.
- The absorbed solar irradiation in the layers beneath the PV cells can be neglected [19,26,30].

2.3. Governing equations

Single-phase and multi-phase models (e.g., the mixture model) have widely been used for nanofluids [31–34]. Salari et al. [32] elucidated that the mixture model is particularly suitable for a nanofluid-based PV/JIC system which is also adopted in this study. Kumar and Puranik [35] further showed that the drag and gravity forces are dominant in turbulent nanofluids, while other forces are negligible. The mixture model solves continuity, momentum, and



(a) Cross-sectional view and the thermal network of the PV/JIC system



(b) Isometric view

Fig. 1. Schematic diagram of the PV/JIC system.

Table 1
Different parameters of solid components in the PV/JIC system [18,19].

	Glass	PV cells	Tedlar	Substrate
Thickness, t, (mm)	4	0.3	1	0.5
Thermal conductivity, k, (W/mk)	1	148	0.033	202
Absorptivity, α	0.05	0.9	–	–
Transmissivity, τ	0.92	0.09	–	–
Emissivity, ε	0.85	–	–	–

Table 2
Thermophysical properties of water and SiC particles [22].

Material	Density (kg/m^3)	Thermal conductivity (W/m·K)	Specific heat capacity (J/kg·K)
Water	998.2	0.6	4182
SiC	3220	490	737

energy conservation equations in the fluid domain which are expressed by the following equations [31,32,36–38].

- Continuity equation

$$\frac{\partial(\rho_m)}{\partial t}+\nabla.\left(\rho_m \vec{v}_m\right)=0 \tag{1}$$

where $\vec{v}_m$ and ρ_m are the mass-averaged mixture velocity and the mixture density, respectively, calculated as

$$\vec{v}_m=\frac{\sum_{k=1}^{n}\varphi_k\rho_k\vec{v}_k}{\rho_m} \tag{2}$$

$$\rho_m=\sum_{k=1}^{n}\varphi_k\rho_k \tag{3}$$

where φ_k is the volume fraction of phase k, and n is the number of phases.

- Momentum equation

$$\frac{\partial\left(\rho_m\vec{v}_m\right)}{\partial t}+\nabla.\left(\rho_m\vec{v}_m\vec{v}_m\right)=-\nabla P+\nabla.\left[\mu_m\left(\nabla\vec{v}_m+\nabla\vec{v}_m^{T}\right)\right]+\nabla.\left(\sum_{k=1}^{n}\varphi_k\rho_k\vec{v}_{dr,k}\vec{v}_{dr,k}\right)+\rho_m\vec{g}+\vec{F} \tag{4}$$

where $\vec{F}$ is the body force. $\vec{v}_{dr,k}$ represents the drift velocity of secondary phase k, defined as [33].

$$\vec{v}_{dr,k}=\vec{v}_k-\vec{v}_m \tag{5}$$

And the mixture viscosity (μ_m) is expressed as follows

$$\mu_m=\sum_{k=1}^{n}\varphi_k\mu_k \tag{6}$$

where μ_k is the dynamic viscosity of phase k. It is noted that the effective solid viscosity of nanoparticles (μ_{np}) is calculated as [39,40].

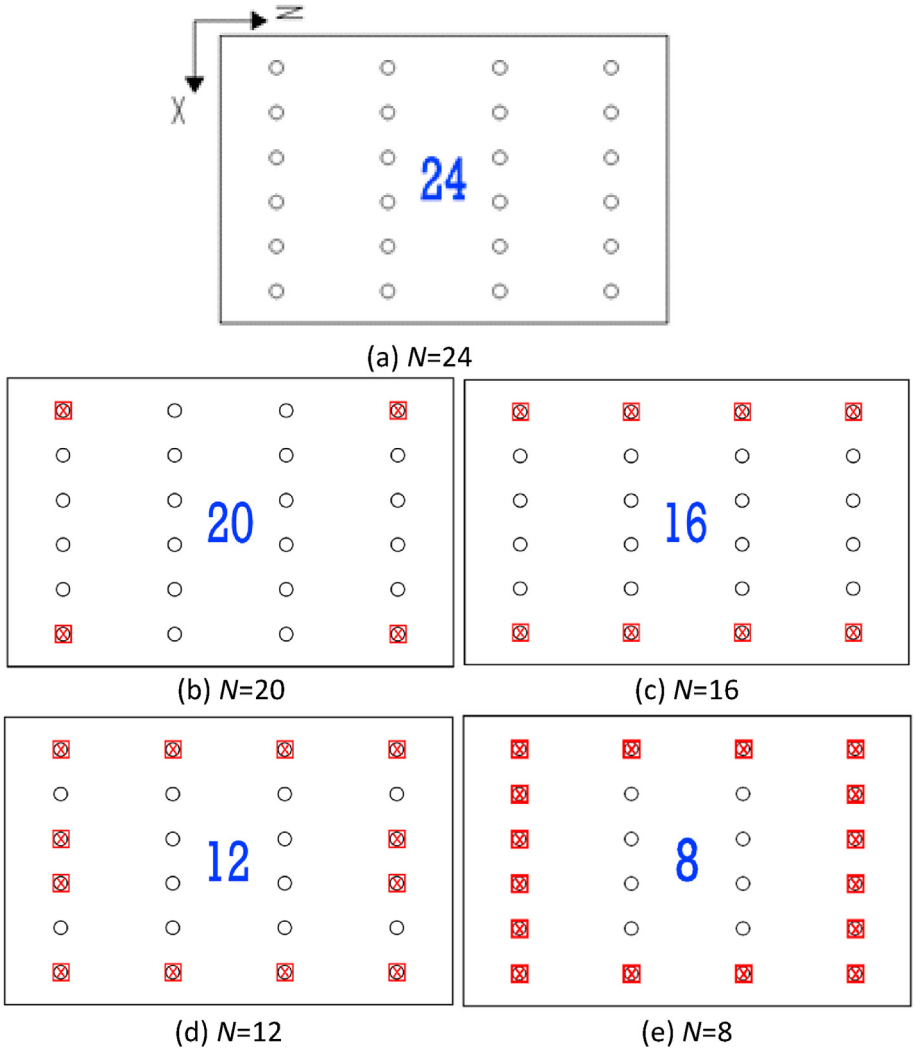


Fig. 2. In-line configuration of nozzle arrangement on the JIC system.

$$\mu_{np} = -0.188 + 537.42\varphi_{np} \quad (7)$$

- Energy equation

$$\frac{\partial \sum_{k=1}^{n}\left(\varphi_k \rho_k c_{p,k} T_k\right)}{\partial t} + \nabla.\sum_{k=1}^{n}\left(\varphi_k \rho_k c_{p,k} \vec{v}_k T_k\right) = \nabla.\left(k_{eff}\nabla T\right) \quad (8)$$

where c_p is the specific heat capacity, and k_{eff} is the mixture conductivity defined as [33].

$$k_{eff} = \sum_{k=1}^{n} \varphi_k (k_k + k_t) \quad (9)$$

where k_t is the turbulent thermal conductivity.

- Volume fraction equation.

The volume fraction of the secondary phase p is expressed as [33].

$$\frac{\partial}{\partial t}(\varphi_p \rho_p) + \nabla.\left(\varphi_p \rho_p \vec{v}_m\right) = -\nabla.\left(\varphi_p \rho_p \vec{v}_{dr,p}\right) \quad (10)$$

where drift velocity ($\vec{v}_{dr,p}$) is expressed as follows [33].

$$\vec{v}_{dr,p} = \vec{v}_{pq} - \sum_{k=1}^{n} \frac{\varphi_k \rho_k}{\rho_m} \vec{v}_{qk} \quad (11)$$

where $\vec{v}_{pq}$ is the relative velocity.

The renormalization group (RNG) k-ε model is adopted here as it has widely been used for predicting complex flow and thermal fields. It is superior to other turbulence models in simulating multiphase impingement jet flows [41] due to its accuracy and convergence speed [42–44]. The RNG k-ε turbulence model solves the flow and thermal fields [45] using the following equations:

$$\frac{\partial(\rho k)}{\partial t} + \nabla.(\rho k v) = \nabla.\left\{\left(\mu + \frac{\mu_t}{\sigma_k}\right)\nabla k\right\} + P_k + P_b - \rho\varepsilon \quad (12)$$

where P_k and P_b are the turbulent kinetic energy production terms by the shear strain rate and buoyancy effects, respectively.

$$\frac{\partial(\rho\varepsilon)}{\partial t} + \nabla.(\rho\varepsilon v) = \nabla.\left\{\left(\mu + \frac{\mu_t}{\sigma_\varepsilon}\right)\nabla\varepsilon\right\} + \frac{\varepsilon}{k}(C_{1\varepsilon}P_k - C_{2\varepsilon}\rho\varepsilon) \quad (13)$$

where

$$C_{1\varepsilon} = 1.42 - \frac{\eta\left(1 - \eta/\eta_0\right)}{1 + \beta\eta^3} \quad (14)$$

$$\eta = \sqrt{\frac{P_k}{\rho C_\mu \varepsilon}} \quad (15)$$

where $C_\mu = 0.0845$.

2.4. Energy balance equations of the PV system

Heat dissipation due to radiation and convection is included for the top surface of the glass layer [22]. The energy balance equation in the glass component is expressed as.

$$I\alpha_g = h_w^*(T_g - T_{amb}) + \sigma\varepsilon\left(T_g^4 - T_{sky}^4\right) + k_{g-pv}(T_g - T_{pv}) \quad (16)$$

where σ is the Stefan-Boltzmann constant. h_w^* and T_{sky} represent the heat transfer coefficient and the sky temperature, respectively. They are defined as [22,46]:

$$h_w^* = \begin{cases} V_w^{0.78} + 6.47 if V_w > 5\ m/s \\ 3.8V_w + 5.7 if V_w < 5\ m/s \end{cases} \quad (17)$$

$$T_{sky} = 0.0522 T_{amb}^{1.5} \quad (18)$$

where V_w is the wind speed and equals 1 m/s [22].

A slight portion of solar irradiation in the PV layer is converted to electrical output. The following equation calculates the PV efficiency [2].

$$\eta_{PV} = \eta_{ref}\left[1 - 0.0045\left(T_{cell} - T_{ref}\right)\right] \quad (19)$$

where T_{cell} is the average temperature of PV cells, T_{ref} is the reference temperature (25 °C), and η_{ref} refers to the electrical module efficiency under the standard test condition, which is equal to 16.5% based on the PV datasheet.

The energy balance equation of the PV component is [15]:

$$I\tau_g\alpha_{pv}(1 - F\eta_{PV}) + k_{g-pv}(T_g - T_{pv}) = k_{pv-tedlar}(T_{pv} - T_{tedlar}) \quad (20)$$

where F is the packing factor of the PV panel ($F = 0.9$) [47,48].

The energy balance equations in the tedlar backsheet and aluminum substrate layers are expressed as follows.

$$k_{pv-tedlar}(T_{pv} - T_{tedlar}) = k_{tedlar-substrate}(T_{tedlar} - T_{substrate}) \quad (21)$$

$$k_{tedlar-substrate}(T_{tedlar} - T_{substrate}) = k_{substrate-fluid}\left(T_{substrate} - T_{fluid}\right) + \dot{m}c_{p,fluid}\Delta T_{fluid} \quad (22)$$

where the fluid subscript refers to nanofluids and the air.

2.5. Boundary conditions

Since natural sunlight is not constantly accessible, some studies have considered a sunlight simulator to provide adjustable and uniform irradiation [2,22,49]. Five arrangements of nozzles are studied, as shown in Fig. 2. The ambient temperature (T_{amb}) is set to 25 °C. SiC-water nanofluids with a concentration of 4% enter the domain at four mass flow rates (0.045, 0.07, 0.095, and 0.12 kg/s) with different inlet temperatures (20, 27, 34, 40 °C) [2,16,22]. The inlet velocity (V_{jet}) is calculated as

$$V_{jet} = \frac{\dot{m}}{\rho N\left(\frac{\pi}{4}d_n^2\right)} \quad (23)$$

where N is the number of nozzles, d_n is the nozzle diameter.

Boundary conditions at the inlets are expressed as follows.

$$v = V_{jet},\ u = w = 0,\ T_f = T_{jet},\ \varphi_{water} = 0.96\ and\ \varphi_{SiC} = 0.04 \quad (24)$$

The open boundary condition is applied for outlets as [50].

$$P = P_{atm} \tag{25}$$

$$\frac{\partial u}{\partial n} = \frac{\partial v}{\partial n} = \frac{\partial w}{\partial n} = \frac{\partial T_f}{\partial n} = 0 \tag{26}$$

where n refers to outward drawn normal. Due to the symmetry of fluid flow and heat transfer, one-quarter of the domain is numerically analysed to save computational resources and time. Hence, the symmetry conditions are applied on the X-axis and Z-axis sides.

There is no heat loss at the side walls of the PV components and hence:

$$\frac{\partial T_s}{\partial n} = 0 \tag{27}$$

2.6. Numerical setup

In this study, the governing equations are discretised using a pressure-based solver that uses the finite volume method (FVM). Velocity is coupled with pressure by the SIMPLE pressure correction algorithm. The compressive scheme is adopted for the volume fraction [51], whereas the second order upwind scheme is considered for other parameters [22]. All simulations are carried out by ANSYS Fluent 2021 R2. The mixture model is used for solving the fluid domain, including three phases (air, water, and solid nanoparticles). All simulations with/without cooling are conducted in transient (up to 80 min) or pseudo transient states to reach steady conditions [2].

2.7. Mesh study

Fig. 3 presents a 3D mesh grid generated for fluid and solid zones. The mesh refinement is implemented with acceptable orthogonality near the impingement surface to capture the effect of hydrodynamic and thermal boundary layers ($Y^+<1$).

A mesh independence study is carried out on the average temperature over the target surface for four different meshes (900, 1500, 2700, and 4400 K) when the mass flow rate of water is 0.045 kg/s. As shown in Table 3, the mesh resolution initially affects the prediction of the average temperature, while further refinement has a negligible impact on the result for the cell numbers higher than 2700 K. Hence, the case with 2,700,000 elements is selected for simulations in this study.

2.8. Numerical validation

The validation of the present numerical approach is conducted in two parts: (a) the mixture model for nanofluids and (b) the PV/JIC system. The mixture model validation case includes a microchannel with a hydraulic diameter of 341 μm by 4.48 cm long, which is simulated under the heat flux of 300 W. Nanofluids with the inlet temperature of 30 °C enter the domain at four different mass flow rates. The results are presented in Fig. 4(a), confirming excellent compatibility between predicted results and existing data [52]. The maximum deviation of 7.76% is observed for the temperature difference between the inlet and outlet, which is consistent with our previous study [36]. For the PV/JIC system, the present model is validated against the experiment of Javidan and Jabari [2]. For the validation, the coolant is pure water with an inlet temperature of 18 C. At $N = 8$, $d_n = 2$ mm, and $H/d_n = 15$, the mass flow rate varies from 0.02 to 0.12 kg/s. As can be seen in Fig. 4(b), the model captures the decreasing trend very well, although it slightly underpredicts the temperature values. To further explore the accuracy of the model, in addition to RNG k-ε, the simulations are conducted using the SST k-ω which is also a popular turbulence model [42,43,53]. However, the results confirm that the RNG k-ε model performs better which are presented here. The observed discrepancy could also be due to uncertainties in measured data. An uncertainty of 4.21% was reported in measuring the average surface temperature in the experimental study [2], which could be due to the number and accuracy of thermocouples in measuring temperatures. Nevertheless, the agreement is acceptable, showing a reasonable deviation (11%–15%) from the measured data.

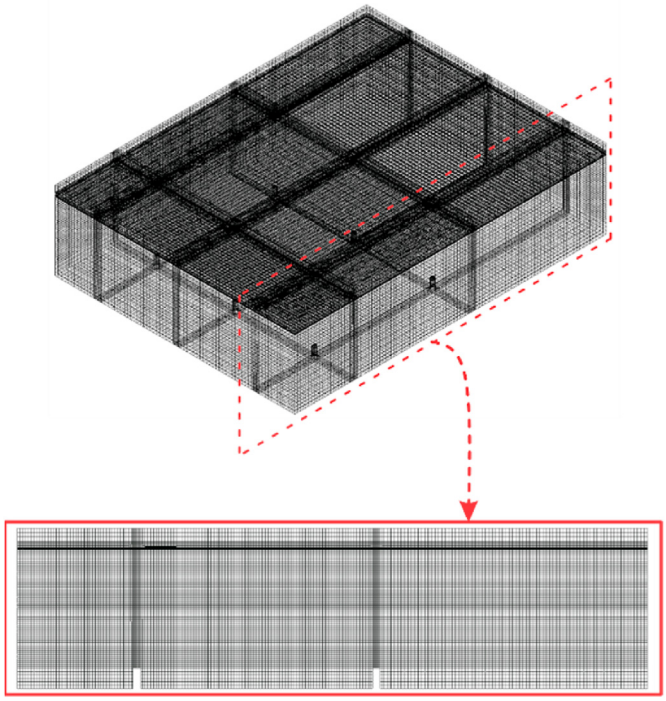


Fig. 3. Mesh configuration of the computational domain.

Table 3
A mesh independence study at $\dot{m} = 0.045$ kg/s, $\varphi = 0\%$, $H = 30$ mm, $N = 8$, $d_n = 2$ mm, and $T_{in} = 20$ C.

Mesh	Mesh number	The mean PV temperature (°C)
Coarse	900,000	40.05
Medium	1,500,00	41.83
Fine	2,700,000	42.14
Finest	4,400,000	42.32

3. Result and discussion

This section discusses the impact of different parameters on flow structure and cooling performance.

3.1. Flow field structure

As noted previously, jet impingement is an efficient cooling method that provides intensive and localized heat removal from a hot surface. A thin thermal boundary layer forming over the target surface controls the heat transfer rate in this method. The flow structure of jet impingement is complex due to high-velocity gradients resulting in higher turbulence and heat transfer [43,54]. Fig. 5 sheds more light on the flow structure of a free impinging jet

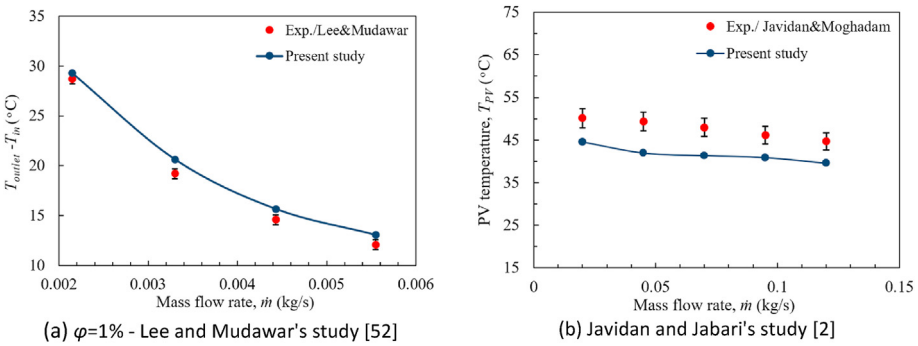


Fig. 4. Validation of the numerical model with the experimental data [2,52].

[54]. In the potential core (Region R–I), the jet velocity remains constant, while dissipating the axial velocity and spreading the established flow in the transverse direction can be seen in Region R–II. The axial jet velocity is deflected in the stagnation zone (Region R–III). The thermal and momentum exchange rates are highly dependent on the potential flow over the boundary layers. This zone provides the maximum heat transfer. In the wall jet region (Region R–IV), the directed velocity of the bulk flow enhances in thickness when the boundary layer forms over the target surface.

Multiple jets can enhance temperature uniformity, giving a higher heat transfer coefficient. Fig. 6 shows two possible interactions in multiple jets: (i) the interactions between adjacent IJs affecting the impinging intensity in stagnation zones, and (ii) the interactions of IJs with the flow formed by neighbouring jets on the target surface, which impact flow characteristics in wall jet regions [43]. As a result, the flow structure and heat transfer of multiple IJs can be slightly different from a single jet, shown in Fig. 5 [43,55]. The dimensionless velocity profile of active jets in the XY-plane (Fig. 6(b)) shows that the thickness of the hydrodynamic boundary layer in wall jet regions can be affected by moving flows from stagnation zones on the target surface. This enhances the heat transfer since reducing the boundary layer results in higher heat transfer [42,43], which will be discussed further in the following sections.

3.2. Comparing the PV/JIC system with uncooled PV

Fig. 7 compares the evolution of temporal mean temperature over the PV panel for three different scenarios. The mean temperature of the PV system without the cooling system reaches 68.5 °C which is significantly higher than the PV/JIC systems. However, integrating a cooling system with the PV module results in lower temperature distribution. The effect of the cooling system on the PV module, including pure water and nanofluids, is also shown for eight nozzles at $H = 30$ mm impinging on the target surface. The surface temperature drops to 42.1 °C using water jet impingement. Further heat transfer enhancement can be achieved by adding SiC nanoparticles. It can be seen that the dispersion of SiC nanoparticles into the water reduces the surface temperature by 27.4 °C, which is due to their potential to absorb a large portion of the heat

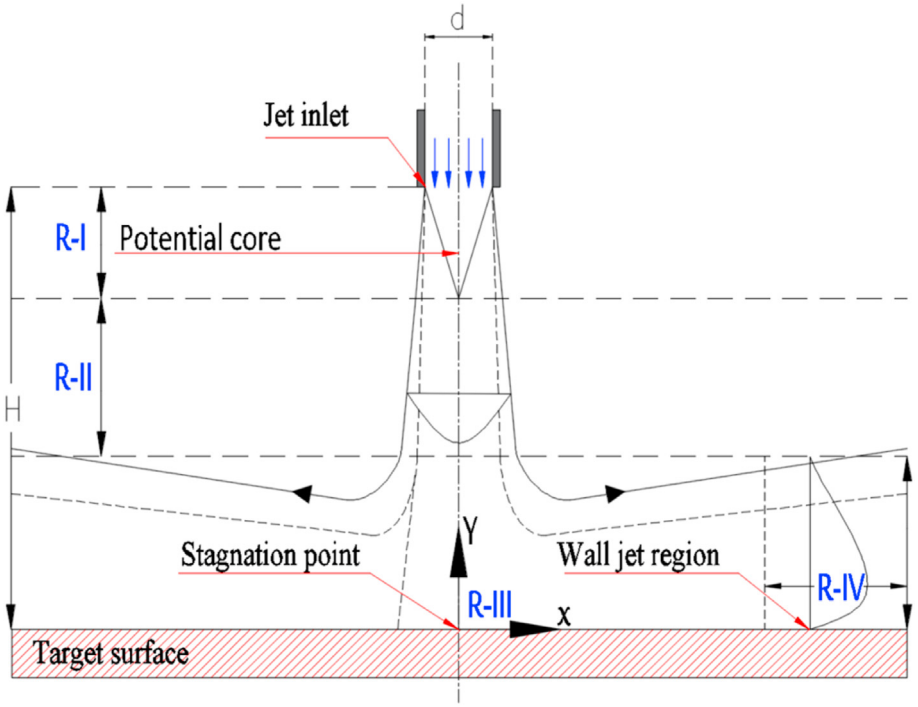
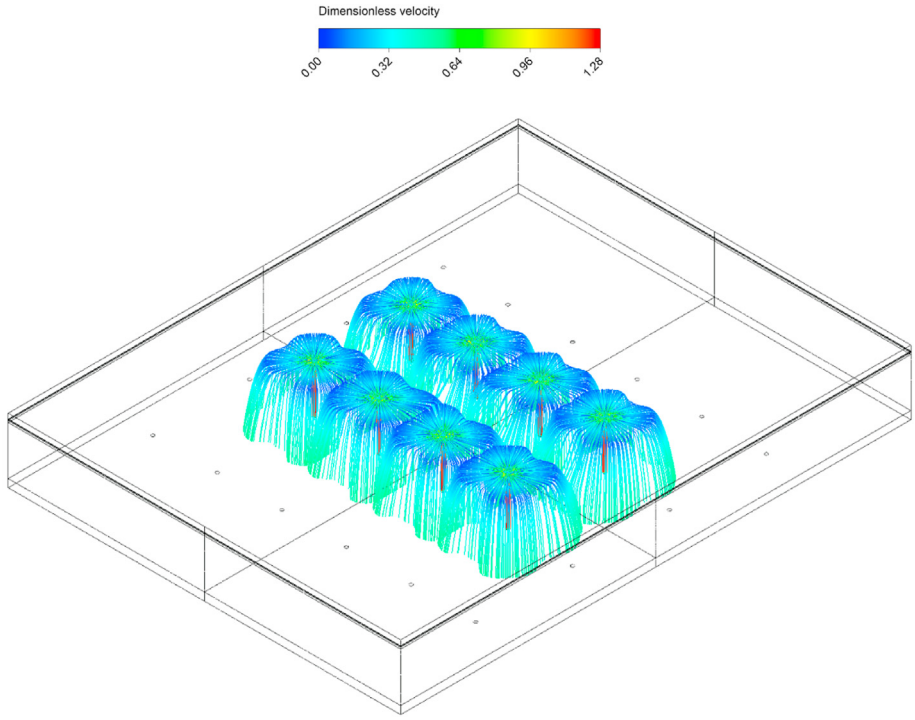


Fig. 5. Jet impingement flow structure [54].



(a) 3D Streamlines

(b) Velocity profile in the XY-plane

Fig. 6. Flow structure of the PV/JIC system at $\dot{m} = 0.045$ kg/s, $\varphi = 4\%$, $H = 30$ mm, $N = 8$, and $d_n = 2$ mm.

[16]. SiC particles increase the thermal conductivity of the mixture, and hence nanofluid jet impingement improves PV cooling performance better than water.

3.3. Effect of the inlet temperature

The inlet temperature of coolants highly impacts the efficiency of the PV/JIC system [24]. Lower inlet temperatures can transfer higher heat from PV cells, resulting in higher efficiency [24]. Here, the JIC system injects SiC nanofluids at the mass flow rate of 0.045 kg/s from 8 nozzles located at 30 mm from the rear surface of the PV module. Fig. 8 shows the effect of the inlet temperature of nanofluids on the PV efficiency obtained using Equation (19). It can be seen that increasing the inlet temperature reduces the PV efficiency, which is consistent with previous studies [24]. The maximum PV efficiency of 15.3% is obtained when the inlet temperature of SiC-water nanofluids is 20 °C.

3.4. Effect of the mass flow rate

Temperature uniformity and heat transfer enhancement are two significant parameters that should be considered to design a cooling system [43]. The temperature uniformity index (T') is defined to measure the uniformity of PV temperature over the surface. It is noted that the lower value of the temperature uniformity index shows more uniformity. This area-weighted index is calculated as follows.

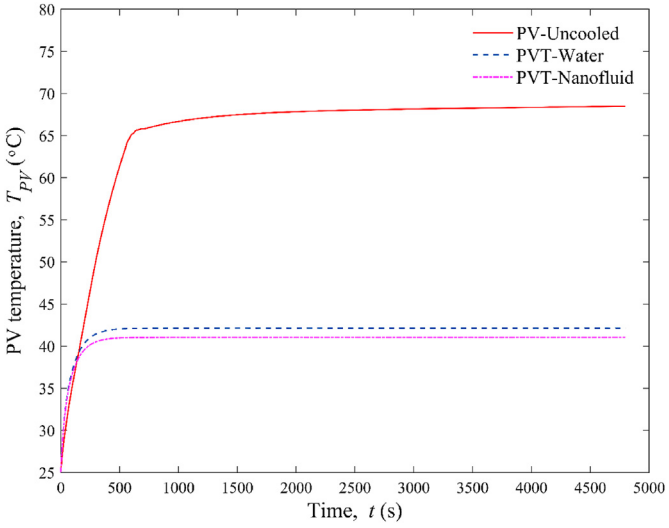


Fig. 7. Comparing the efficiency of the uncooled PV to the PV/JIC system with water/nanofluids at $\dot{m} = 0.045$ kg/s, $H = 30$ mm, $N = 8$, $d_n = 2$ mm, and $T_{in} = 20$ °C.

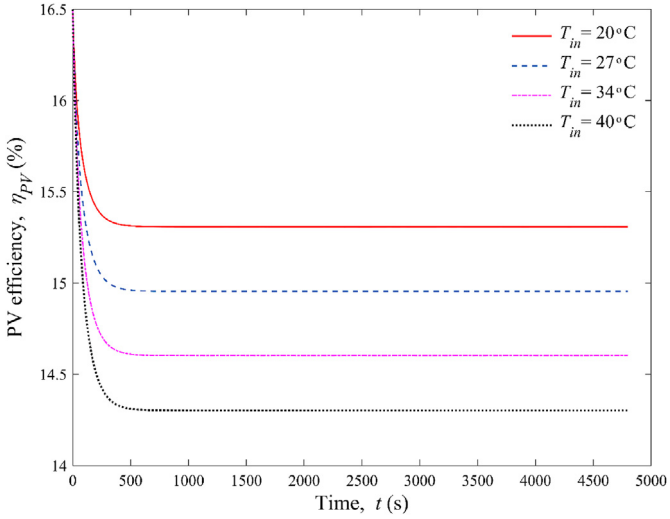


Fig. 8. Effect of the inlet temperature of SiC-water nanofluids on the PV efficiency at $\dot{m}$ = 0.045 kg/s, φ = 4%, H = 30 mm, N = 8, and d_n = 2 mm.

$$T' = \frac{\sum_{i=1}^{n}\left[\left(\left|T_i - T_{avg}\right|\right)\right]A_i}{\sum_{i=1}^{n}A_i} \tag{28}$$

where T_{avg} is the average temperature, and n represents the number of elements.

Fig. 9 shows the PV temperature and the temperature uniformity for different mass flow rates in an arrangement of 8 nozzles at H = 30 mm when the inlet temperature is 20 °C. Increasing the nanofluid mass flow rate boosts the capability of the JIC system to provide further uniform temperature distribution. The minimum temperature uniformity index (T') of 0.36 °C is obtained at $\dot{m}$ = 0.12 kg/s. However, there is an inverse proportion between the mass flow rate and PV temperature. Increasing the mass flow rate from 0.045 to 0.12 kg/s reduces the mean PV temperature by 4.6 °C under a solar irradiance of 1000 W/m^2. This can be due to reducing the thickness of hydrodynamic and thermal boundary layers over the target surface caused by increasing jet velocity magnitude at the inlet [43].

In addition, the efficiency enhancement of PV/JIC compared to the PV module is presented in Fig. 10. Higher mass flow rates of SiC-water nanofluids result in higher PV efficiencies, and rising the mass flow rate from 0.045 to 0.12 kg/s improves the PV by 17.93%.

To shed more light on the effect of different mass flow rates on the flow structure, Fig. 11 shows the turbulent intensity variation against streamwise distance at z = 128 mm on the target surface. Increasing the mass flow rate enhances the momentum exchange rate near the wall. Further, velocity augmentation results in moving more vigorous vortices from stagnation zones toward wall jet regions. As a result, turbulent intensity dominantly increases in these regions. Therefore, higher turbulent intensity enhances the heat transfer rate remarkably.

3.5. Effect of nozzle number

The effect of the nozzle number at the constant mass flow rate is analysed for the nanofluid JIC system. Previous studies have shown that it plays a significant role in heat transfer enhancement and more uniform temperature distribution [2,43].

According to Section 3.4, the maximum efficiency is obtained at the mass flow rate of 0.12 kg/s. Therefore, Fig. 12 demonstrates the PV efficiency improvement against five sets of IJs at $\dot{m}$ = 0.12 kg/s. It can be seen that increasing the number of nozzles from 8 to 12 improves the PV efficiency by 1.42%, while further nozzles show a slight increase. The PV/JIC system enhances the conventional PV efficiency by 15.89% at N = 24 under the same conditions.

To further analysis, the temperature distribution of the minimum and the maximum number of nozzles is depicted in Fig. 13. The maximum surface temperature decreases from 42 to 35.4 °C as the number of nozzles increases from 8 to 24. In addition, the temperature distribution uniformity is improved by this increase.

3.6. Effect of jet-to-surface distance

One of the significant parameters in jet impingement is the jet-to-surface distance [2]. In this section, three different heights (5.1, 30, and 55 mm) are chosen based on the previous studies [2]. The variation of PV temperature and its uniformity is depicted in Fig. 14 against the jet-to-surface span. It can be seen that decreasing the jet-to-surface distance reduces the temperature uniformity index (T') and consequently improves the uniformity while it decreases the surface temperature remarkably. At H = 5.1 mm, the T_{PV} and T'

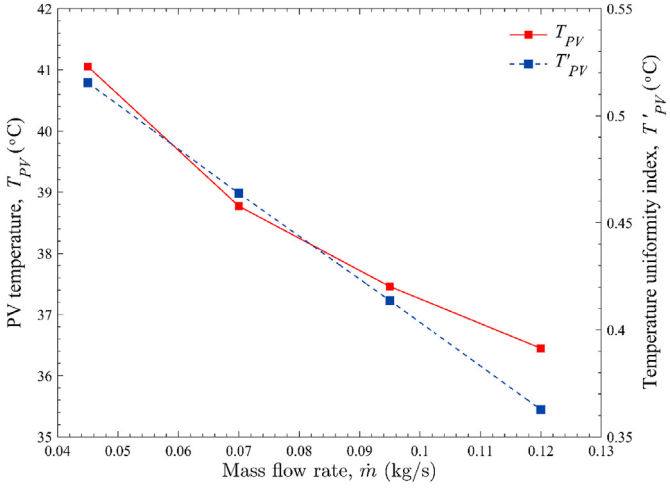


Fig. 9. Effect of different nanofluid mass flow rates on the PV temperature and uniformity at φ = 4%, H = 30 mm, N = 8, d_n = 2 mm, and T_{in} = 20 °C.

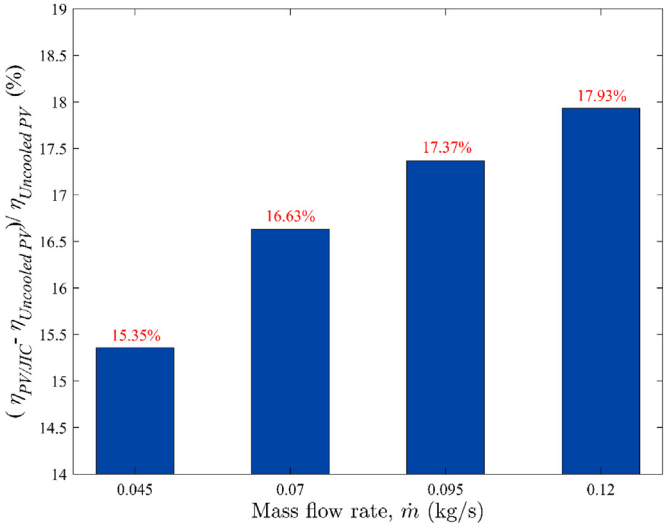


Fig. 10. Effect of different nanofluid mass flow rates on PV efficiency at φ = 4%, H = 30 mm, N = 8, d_n = 2 mm, and T_{in} = 20 °C.

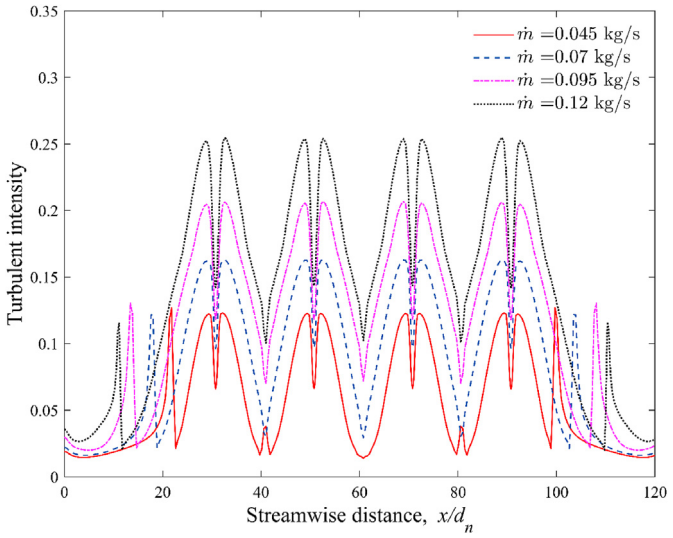


Fig. 11. Turbulent intensity profiles of different mass flow rates at $\varphi = 4\%$, $H = 30$ mm, $N = 8$, $d_n = 2$ mm, and $z = 128$ mm.

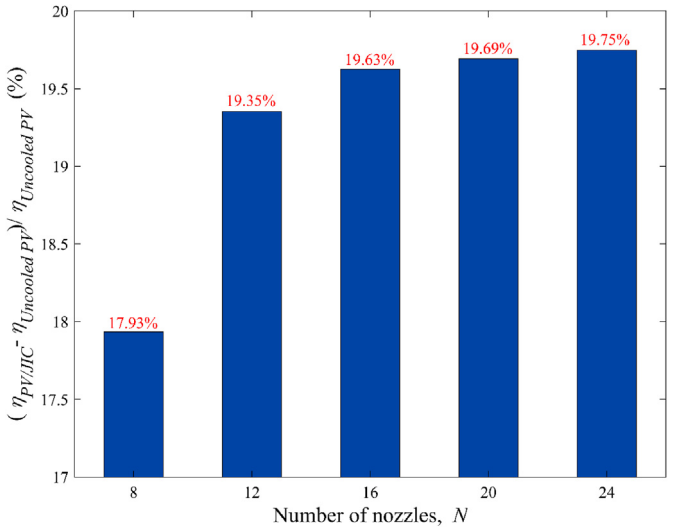


Fig. 12. Effect of different nozzle numbers on the PV efficiency at $\dot{m} = 0.12$ kg/s, $\varphi = 4\%$, $H = 30$ mm, $d_n = 2$ mm, and $T_{in} = 20$ °C.

are minimised to 32.4 °C and 0.019 °C, respectively.

For further investigation, Fig. 15 reveals that when the jet-to-surface distance decreases in the JIC system, larger turbulence kinetic energy is observed downstream of IJs at the wall jet regions and further distances. As a result, rising the turbulence kinetic energy enhances heat transfer and cooling performance in a broader area. Therefore, decreasing the jet-to-surface spacing results in a lower PV temperature and better temperature uniformity, simultaneously, at $H = 5.1$ mm (See Fig. 14). In contrast, increasing the jet-to-surface span leads to the radial dispersion of injected nanofluids at a larger distance from the rear surface of the PV module (see Fig. 15(c)). Hence, the axial impingement velocity decreases on the target surface.

3.7. Effect of the nozzle diameter

The heat transfer augmentation is associated with adverse impacts of higher pressure loss and consequently higher pumping power. In this section, to provide a better overview of the effectiveness of the cooling system, in addition to temperature, the maximum power increment percentage and the *COE* are analysed.

The JIC as an active technique requires the pumping power, calculated as follows [2].

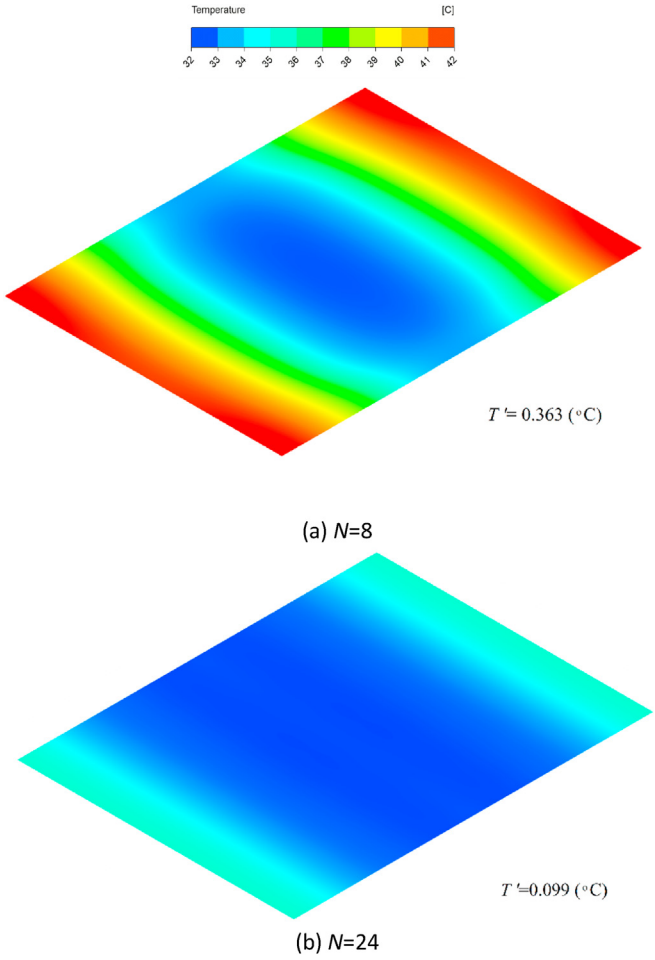


Fig. 13. Temperature distribution contours on the PV panel at $\dot{m} = 0.12$ kg/s, $\varphi = 4\%$, $H = 30$ mm, $d_n = 2$ mm, and $T_{in} = 20$ °C.

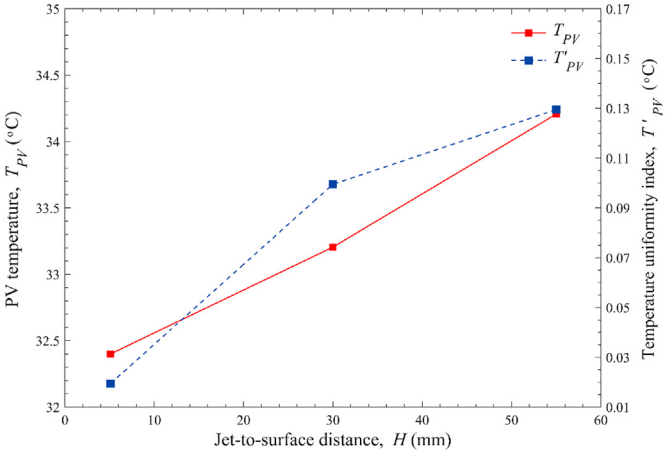


Fig. 14. Effect of different jet-to-surface distances on the PV temperature and uniformity at $\dot{m} = 0.12$ kg/s, $\varphi = 4\%$, $N = 24$, $d_n = 2$ mm, and $T_{in} = 20$ °C.

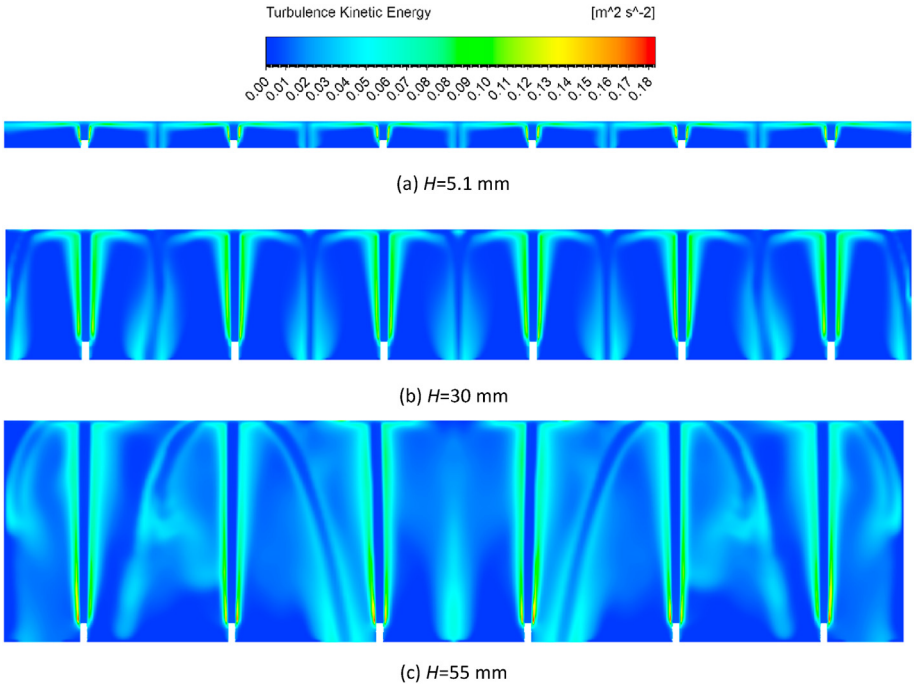


Fig. 15. Turbulence kinetic energy contours in the XY-plane at $\dot{m}$ = 0.12 kg/s, φ = 4%, N = 24, d_n = 2 mm, and T_{in} = 20 °C.

$$P_{Pump} = Q.\Delta P \tag{29}$$

where Q is the volumetric flow rate, and ΔP is the pressure loss.

The maximum power increment percentage is adopted to evaluate the output power of the PV/JIC system in comparison with the conventional PV module, which is defined as [2].

$$\%P_{(\max)increase} = \left(\frac{P_{PV/JIC} - P_{PV,uncooled}}{P_{PV,uncooled}}\right) \tag{30}$$

The *COE* is expressed as the ratio of the net output power of the PV/JIC system to the uncooled PV system [2]. This parameter is employed to determine the economic efficiency of the proposed PV/JIC system.

$$COE = \left(\frac{P_{PV/JIC} - P_{Pump}}{P_{PV,uncooled}}\right) \tag{31}$$

The effect of the nozzle diameter is examined while other parameters are considered as $\dot{m}$ = 0.12 kg/s, N = 24, φ = 4%, H = 5.1 mm, and T_{in} = 20 °C, which represent the most efficient JIC system examined in the previous sections. Different design parameters, including the mean PV temperature, $\%P_{(max)increase}$, *COE*, PV efficiency, are compared for two different nozzle diameters in Table 4. As can be seen, decreasing the nozzle diameter from 2 to 1 mm slightly enhances the mean PV temperature.

Referring to Fig. 7, the mean PV temperature reaches 68.5 °C in the PV module in the abscense of the cooling system. However, this parameter drops by 36.4 °C using the JIC system when $\dot{m}$ = 0.12 kg/s, N = 24, φ = 4%, H = 5.1 mm, d_n = 1 mm, and T_{in} = 20 C.

Table 4
The effect of the nozzle diameter on the performance of the PV/JIC system. at $\dot{m}$ = 0.12 kg/s, N = 24, φ = 4%, H = 5.1 mm, and T_{in} = 20 °C.

Case	d_n (mm)	T_{avg} (C)	$\%P_{(max)increase}$	*COE*	η_{PV} (%)
1	1	32.1	20.36	1.15	15.97
2	2	32.4	20.19	1.18	15.95

However, integrating the PV module with the JIC comes with additional costs. Meanwhile, Table 4 compares the maximum power increment of the PV/JIC system for two cases against estimated costs. It can be concluded that the JIC configuration with a nozzle diameter of 1 mm enhances the maximum output power of the PV/JIC system by 20.36% compared to the conventional PV module. However, in terms of the economic aspect, decreasing the nozzle diameter to 1 mm increases the pumping power, consequently reducing the *COE* to 1.15, which is consistent with previous studies [2].

4. Conclusion

A computational study has been successfully performed by integrating a nanofluid jet impingement cooling system into a real PV solar panel. A three-phase mixture model was used to examine significant parameters affecting PV efficiency, including the inlet temperature number of nozzles, mass flow rate, jet-to-surface spacing, and nozzle diameter.

Numerical results demonstrated nanofluids were superior to pure water in cooling the PV module. The inlet temperature of nanofluids significantly affected the PV temperature, so a lower inlet temperature caused higher output power. In addition, the largest mass flow rate ($\dot{m}$ = 0.12 kg/s) improved PV efficiency and temperature uniformity remarkably. Increasing the number of nozzles to 24 enhanced the PV efficiency by 19.75% and reduced the temperature uniformity index (T') to 0.099 °C. A decrease in jet-to-surface spacing from 55 to 5.1 mm reduced the mean PV temperature by approximately 2 °C.

Regarding the nozzle diameter, along with the thermal and electrical study of the PV/JIC system, the economic aspect of the output power enhancement was deeply analysed. Although decreasing the nozzle size improved the maximum output power of the PV/JIC system, it increased the *COE* to 1.18.

The PV/JIC system with the configuration of $\dot{m}$ = 0.12 kg/s, N = 24, φ = 4%, H = 5.1 mm, d_n = 1 mm, and T_{in} = 20 °C reduced the

mean PV surface temperature to 32.1 °C, while it reached 68.5 °C in the case of the uncooled PV module.

CRediT authorship contribution statement

Javad Mohammadpour: Formal analysis, Methodology, Visualization, Writing – original draft. **Fatemeh Salehi:** Conceptualization, Supervision, Writing – review & editing, Resources. **Mohsen Sheikholeslami:** Conceptualization, Supervision, Writing – review & editing. **Ann Lee:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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